Grade: Check Minus

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Homework #2

1.

x	f(x)	xf(x)	x-u	(x-u)^2	(x-u)^2 f(x)
20	0.05	1.00	-5.41	29.27	1.46
21	0.06	1.26	-4.41	19.45	1.17
22	0.07	1.54	-3.41	11.63	0.81
23	0.08	1.84	-2.41	5.81	0.46
24	0.09	2.16	-1.41	1.99	0.18
25	0.12	3.00	-0.41	0.17	0.02
26	0.15	3.90	0.59	0.35	0.05
27	0.13	3.51	1.59	2.53	0.33
28	0.1	2.80	2.59	6.71	0.67
29	0.1	2.90	3.59	12.89	1.29
30	0.05	1.50	4.59	21.07	1.05

✓ Q1

$\mu=\sum xf(x)$

$\mu=25.41$

$\sigma^2=\sum (x-\mu)^2 f(x)$

$\sigma^2=7.52$

$\sigma=2.74$

2a.

✓ Q2a

$P(\mu-\sigma \leq x \leq \mu+\sigma)$

$P(22.67 \leq x \leq 28.15)$

$P(23 \leq x \leq 28) = f(23)+f(24)+f(25)+f(26)+f(27)+f(28) =0.67$

2b.

✓ Q2b

$P(\mu-2\sigma \leq x \leq \mu+2\sigma)$

$P(19.9 \leq x \leq 30.9)$

$P(20 \leq x \leq 30) = 1.0$

3. v=option value

x	f(x)	Option value based on stock price $25
20	0.05	0
21	0.06	0
22	0.07	0
23	0.08	0
24	0.09	0
25	0.12	0
26	0.15	1
27	0.13	2
28	0.1	3
29	0.1	4
30	0.05	5

Probability distribution of v

v	f(v)	✓ Q3 vf(v)	(v-u)^2	(v-u)^2 f(v)
0	0.47	0	1.85	0.87
1	0.15	0.15	1.46	0.22
2	0.13	0.26	1.21	0.16
3	0.1	0.3	1.12	0.11
4	0.1	0.4	0.92	0.09
5	0.05	0.25	1.23	0.06

x Q4: Needs revision.

4. P(v>0)
P= 1-f(0)
P=1-0.53 = 0.47

5.
$\mu=\sum vf(v)$
$\mu=1.36$

x Q5: Needs revision.

$\sigma^2 =\sum (v-\mu)^2 f(v)$
$\sigma^2 = 1.51$
$\sigma = 1.23$